

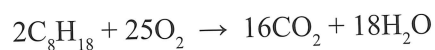
Set 41: Reactions of hydrocarbons

Notes

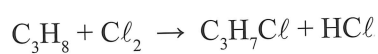
The reactions you will deal with in this section are combustion, substitution and addition reactions.

Examples

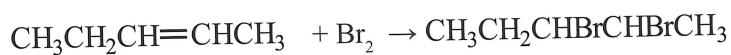
Combustion of hydrocarbons as in the combustion of octane



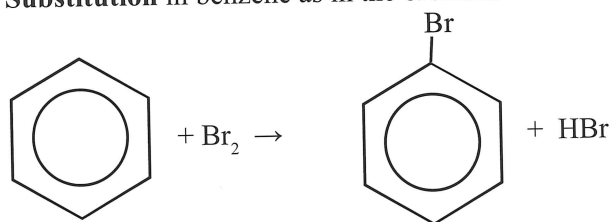
Substitution in alkanes as in the chlorination of propane



Addition in alkenes (and alkynes) as in the bromination of 2-pentene



Substitution in benzene as in the bromination of benzene



Set 41: Exercises

- Hydrocarbons are mostly used as fuels as they all undergo combustion to produce heat. Write equations for reactions between
 - propane and oxygen when ignited.
 - ethene and air when ignited.
 - benzene and air when ignited
 - ethylbenzene and oxygen when ignited



Propane

2. Hydrocarbons are important starting materials for the production of many useful substances. Write equations for and name any organic products formed in the reactions between
- (a) ethane and chlorine in the presence of ultraviolet radiation
 - (b) propene and bromine
 - (c) but-2-ene and hydrogen chloride
 - (d) pent-2-ene and hydrogen in the presence of a platinum catalyst
 - (e) cyclopentene and hydrogen fluoride
 - (f) benzene and bromine
 - (g) propene in excess chlorine
3. As a research chemist you are employed to investigate the conditions that give the best yield of halocarbons starting with simple hydrocarbons. You are required to make the following substances. If there is more than one type of reaction that could produce the product, use the reaction that occurs most readily.
- (i) In each case draw the structure and name the starting hydrocarbon and name any other reagents required.
 - (ii) Write equations for each step of the process.
- (a) chlorofluoromethane
 - (b) chloroethane
 - (c) 1,2-dichloroethane
 - (d) 1,1-dichloroethane
 - (e) 1,1,2-trichloroethane
 - (f) 1-chloro-1-fluoroethane
 - (g) 2-bromo-2-chloropropane
 - (h) iodobenzene
 - (i) chlorobenzene